PEER REVIEW HISTORY

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ARTICLE DETAILS

TITLE (PROVISIONAL)	Effects of Living Kidney Donation on Arterial Stiffness: A Systematic Review Protocol
AUTHORS	Rodriguez, Rosendo; Trentin Sonoda, Mayra; Agharazii, Mohsen; Shorr, Risa; Burns, Kevin

VERSION 1 – REVIEW

REVIEWER	Cenk Gökalp Trakya University Hospital, Turkey
REVIEW RETURNED	12-Nov-2020

GENERAL COMMENTS	I have some advice for the authors about the manuscript: 1- Page 5, Line 6 – please change “live” to “living”, 2- Page 5, Lines 20-27 – please add a reference about a recent study done by Gokalp et al. titled “increase in interventricular septum thickness may be the first sign of cardiovascular change in kidney donors”. 3- Page 8, Lines 41-42 – please change as “intima-media” and “vascular calcification scores”

REVIEWER	Anna Price Queen Elizabeth Hospital Birmingham, United Kingdom
REVIEW RETURNED	07-Dec-2020

GENERAL COMMENTS	The authors have detailed a systemic review protocol to examine the effects of living kidney donation on arterial stiffness. This review is topical, much needed and well overdue. I find it clear and concise. The authors can be commended for a well written protocol on what is challenging topic where studies have much heterogeneity. I have a few comments and suggestions to consider in no particular order: Major comments: 1. Page 8- I note that you are including both those who had an open procedure and laparoscopic nephrectomy. Do you think the surgical approach is important? In theory the haemodynamic stress of an open procedure may be greater and in addition it may take much longer to recover from? Evidence is emerging about weight gain post nephrectomy. Might we expect those that had open procedures to be in more pain and be more sedentary? Do you think this could impact on blood pressure and arterial stiffness measures in donors at least in the short term? I wonder whether as a sub analysis/sensitivity analysis you could ensure there are no significant differences in your results between the approaches?

	2. Page 9 line 11- You are including those with 10 subjects or more. Can you justify this number? This seems exceptionally low when it is considered than many hundreds of patients are required for adequate power for small changes in both pulse wave velocity and change in blood pressure. We are not expecting large differences in blood pressure and arterial stiffness. 3. Will you exclude studies in which measures of arterial stiffness are not adjusted for confounding factors? Without adjusting for heart rate and blood pressure it will be difficult to make any comparisons. Please see the vascular standardisation guidelines below. Perhaps it is worth incorporating this into your interrogation of the data? https://www.ahajournals.org/doi/10.1161/hyp.0000000000000033 4. How will select studies with varying follow up times? I can't see that you are excluding studies with very short follow up. Many studies only follow up patients for several weeks/months after nephrectomy. Will you include these or only those with data of 12 months or more? Minor comments: 1. Page 2 Line 6 "Losing renal mass to unilateral nephrectomy in kidney donors has been associated with an accelerated progression of aortic stiffness and decreased glomerular filtration compared with healthy non-donors." This is a little bold for a starting sentence. I would soften this somewhat. Perhaps 'has been reported to' or 'there is a suggested association'. There are plenty of studies showing no changes! 2. Page 2 Line 32-Do you have the word count to include which metrics. If you could be more specific in the abstract that would be preferable. 3. Page 14 line 10-I agree this sub analysis is really important. 4. Page 17 line 17-I disagree with this statement. I don't think the review will give you more information on 'Long term risks'. Only years of registry studies will enable us to truly know long term risks in donors. We also do not really know what the impact of small changes in arterial stiffness will have on cardiovascular mortality/morbidity, if anything. I suggest you change to 'effects' or 'short to midterm risks'.

VERSION 1 – AUTHOR RESPONSE

Reviewer: 1

Dr. Cenk Gökalp, Trakya University

I have some advice for the authors about the manuscript:
1- Page 5, Line 6 – please change "life" to "living",
We thank the reviewer for this suggestion. This has been changed accordingly on page 5, line 6.

2- Page 5, Lines 20-27 – please add a reference about a recent study done by Gokalp et al. titled "increase in interventricular septum thickness may be the first sign of cardiovascular change in kidney donors".

We appreciate the reviewer's suggestion, this reference has been added to the introduction section, line 10 to 12, page 5.

3- Page 8, Lines 41-42 – please change as “intima-media” and “vascular calcification scores”

We thank the reviewer for this suggestion, we have corrected accordingly on page 7, line 18 and in other locations.

Reviewer: 2
Dr. Anna Price, Queen Elizabeth Hospital Birmingham.

The authors have detailed a systemic review protocol to examine the effects of living kidney donation on arterial stiffness. This review is topical, much needed and well overdue. I find it clear and concise. The authors can be commended for a well written protocol on what is challenging topic where studies have much heterogeneity.

We thank the reviewer for her positive comments and appreciate her feedback on this important clinical issue.

I have a few comments and suggestions to consider in no particular order:

Major comments:

1. Page 8- I note that you are including both those who had an open procedure and laparoscopic nephrectomy. Do you think the surgical approach is important? In theory the haemodynamic stress of an open procedure may be greater and in addition it may take much longer to recover from? Evidence is emerging about weight gain post nephrectomy. Might we expect those that had open procedures to be in more pain and be more sedentary? Do you think this could impact on blood pressure and arterial stiffness measures in donors at least in the short term? I wonder whether as a sub analysis/sensitivity analysis you could ensure there are no significant differences in your results between the approaches?

We thank the reviewer for her insight on this topic. Since its inception (~1995), laparoscopic nephrectomy has been more frequently used worldwide for living donor nephrectomy and some studies have shown its superiority over open nephrectomy. Although open donor nephrectomy is associated with shorter warm ischemia and operative times, donors undergoing laparoscopic nephrectomy may benefit from decreased length of hospitalization, earlier return to employment and less pain without compromising graft function. A recent systematic review and meta-analysis of 13 studies [Am J Transl Res 2020; 12(10): 5993-6002] that compared these 2 procedures in kidney donors showed that serum creatinine levels, rates of hypertension, proteinuria, health-related quality of life and recipient graft survival were not significantly different between these 2 surgical techniques at 1 year after nephrectomy. To our knowledge, there are no reported studies on differences in arterial stiffness related to these procedures. Although experience, training and learning curve for laparoscopic procedures are factors to consider in their effects on the speed of recovery, we agree that if our eligible studies provide sufficient detail regarding the surgical technique, performing a sub-group analysis between these 2 procedures would be desirable to answer this question.

In response to this suggestion, we have added the potential for subgroup analysis on the surgical technique for living donor nephrectomy on page 15, line 15 as follows:

To minimize bias in our review and depending on the number of eligible studies, nature and detail of the reported information, we will attempt to determine the effects of potential confounders by performing sub-group analyses based on age, surgical technique (open vs laparoscopic nephrectomy), duration of follow-up after nephrectomy (short vs long-term), type of comparator

(recipient-related versus non-recipient-related), number and type of medications used for the control of blood pressure and sex according to different male to female ratios.

2. Page 9 line 11- You are including those with 10 subjects or more. Can you justify this number? This seems exceptionally low when it is considered than many hundreds of patients are required for adequate power for small changes in both pulse wave velocity and change in blood pressure. We are not expecting large differences in blood pressure and arterial stiffness.

We thank the reviewer for this suggestion and we have addressed this issue as follows:

Systematic reviews of interventions aim to synthesize all available evidence according to pre-specified eligibility criteria. Unfortunately, such criteria rarely address sample size. Arguments for including small sample size studies uphold that evidence synthesis is best informed by including all reasonably unbiased evidence (including small sample size studies) and that no such evidence should be discarded lightly. Although there is no consensus on a specific cut-off for accepting studies based on sample size, studies with low sample size (i.e. feasibility studies, case-series and pilot trials) are helpful for the following reasons: a) to observe the heterogeneity in the effects across multiple studies; b) to document the heterogeneity that would occur in clinical practice; and c) to identify the intervention effects on relevant sub-groups that would increase generalizability of these results. Moreover, any difficulties that may be caused by including small sample size studies on reporting biases and related small-study effects can certainly be addressed through statistical methods of adjustment and sensitivity analyses. We believe that selecting a low cut-off of 10 participants will allow for a sufficiently broad inclusion of studies to encompass the likely diversity of studies, without loss of information on relevant population sub-groups. This will facilitate the generalizability of our results.

3. Will you exclude studies in which measures of arterial stiffness are not adjusted for confounding factors? Without adjusting for heart rate and blood pressure it will be difficult to make any comparisons. Please see the vascular standardisation guidelines below. Perhaps it is worth incorporating this into your interrogation of the data?
https://www.ahajournals.org/doi/10.1161/hyp.0000000000000033

We thank the reviewer for this suggestion and have addressed this issue as follows:
We agree with the reviewer that the most significant physiologic variable affecting arterial stiffness is systolic blood pressure. In fact, age-related increases in blood pressure are mainly attributable to an increase in systolic blood pressure which leads to a widening in the pulse pressure and a subsequent increase in arterial stiffness. However, the relationship between arterial stiffness and heart rate is less well defined. While our study intends not to exclude arterial stiffness data based on the type of reported values (i.e. adjusted or non-adjusted), we agree with the reviewer that a distinction should be made with regards to how these values are reported. Although AHA recommendations suggest taking in consideration these confounders of arterial stiffness (i.e. blood pressure, heart rate), we speculate that not all studies report adjusted values. To clarify this reporting issue, our study is planning the following: a) to analyze the trends in systolic and diastolic blood pressures as part of our secondary outcomes, so the relationship between blood pressure and metrics of arterial stiffness can be identified with adjusted and non-adjusted values; and b) if feasible, we will perform a stepwise sensitivity analysis with and without adjusted values to identify whether the type of reporting values have an impact on the overall effect estimates. We believe that this approach will address if confounding is real or artifactual.

We have expanded this information in the data extraction section, page 13, lines 8-9 as follows:

We will document if arterial stiffness values are reported as adjusted or non-adjusted according to changes in mean arterial blood pressure and/or heart rate.

Also, we have complemented this information in our sensitivity analysis section, page 15, lines 2-5, as follows:

Depending on the number of eligible studies, sensitivity analyses may be performed to evaluate the effects of extreme values, high risk of study bias, and any confounding effects associated with differences in the reporting of adjusted or non-adjusted values of arterial stiffness according to changes in mean arterial pressure and/or heart rate.

4. How will select studies with varying follow up times? I can't see that you are excluding studies with very short follow up. Many studies only follow up patients for several weeks/months after nephrectomy. Will you include these or only those with data of 12 months or more?

We thank the reviewer for this suggestion. Our review attempts to include all reported follow-up periods as we consider that different time-points of measurement post-nephrectomy may contribute to the heterogeneity of the overall effect estimates. As suggested by the reviewer, we expect that some studies have followed donors for short periods of time (weeks/months) while others have used longer follow-up periods. We believe that including and analyzing this evidence is important for the purposes of tracing the progression of arterial stiffness and for discriminating between short and long-term effects after kidney donation. Depending on the availability and detail of the information, sub-group analysis based on different follow-up periods (short vs long-term) would address the role of time on the effects of the nephrectomy.

We have clarified this issue in our sub-group analysis, page 15, lines 12 to 18, as follows:

To minimize bias in our review and depending on the number of eligible studies, nature and detail of the reported information, we will attempt to determine the effects of potential confounders by performing sub-group analyses based on age, surgical technique (open vs laparoscopic nephrectomy), duration of follow-up after nephrectomy (short vs long-term), type of comparator (recipient-related versus non-recipient-related), number and type of medications used for the control of blood pressure and sex according to different male to female ratios.

Minor comments:
1. Page 2 Line 6 "Losing renal mass to unilateral nephrectomy in kidney donors has been associated with an accelerated progression of aortic stiffness and decreased glomerular filtration compared with healthy non-donors." This is a little bold for a starting sentence. I would soften this somewhat. Perhaps 'has been reported to' or 'there is a suggested association'. There are plenty of studies showing no changes!

Following your suggestion, we have softened our starting sentence in the abstract section (page 2, lines 2-3), as follows:

Introduction: Kidney donors have been reported to have accelerated progression of aortic stiffness and decreased glomerular filtration compared with healthy non-donors.

2. Page 2 Line 32-Do you have the word count to include which metrics. If you could be more specific in the abstract that would be preferable.
Abstract count: 269 words. Text count: 3147 words (including Table).
We have revised the abstract section.

3. Page 14 line 10-I agree this sub analysis is really important.
Thanks for the comment.

4. Page 17 line 17-I disagree with this statement. I don't think the review will give you more information on 'Long term risks'. Only years of registry studies will enable us to truly know long term risks in donors. We also do not really know what the impact of small changes in arterial stiffness will have on cardiovascular mortality/morbidity, if anything. I suggest you change to 'effects' or 'short to midterm risks.'

We agree with the reviewer's statement and so, we have changed our conclusion accordingly as follows (page 18, line 14-16):

Our review will provide a more accurate estimate of the "short-to-mid-term" risks of cardiovascular disease after donation by examining the progression of arterial stiffness post-nephrectomy.

Additional revised changes to the manuscript:

Since progression of arterial stiffness in kidney donors may be a determinant of cardiovascular outcomes, we have revised the objectives section to note that in addition to evaluation of differences in arterial stiffness between donors and healthy controls, we will compare arterial stiffness in kidney donors before and after nephrectomy.

This has been clarified in page 6 lines 21 to 23, of the Objectives section as follows:

We will primarily determine differences in functional markers of arterial stiffness between kidney donors and their respective comparative healthy non-donor controls. Since progression of arterial stiffness in kidney donors may be a determinant of cardiovascular outcomes, we will additionally compare arterial stiffness in donors before and after nephrectomy.

This has also been clarified in page 8, lines 12-14, of the Outcomes section as follows:

The primary outcome will be the difference in the functional metrics of arterial stiffness between healthy nephrectomized and non-nephrectomized participants as well as the difference in donors before and after nephrectomy.

Editorial requirements:

- Please provide the precise dates of your planned systematic search in the abstract and Methods.
Our planned systematic search will proceed until March 2021. We have added this information to the abstract (page 2, lines 10 and 11) and methods section (page 9, line 21) as suggested.

Abstract:
We will comprehensively search for studies published between January 1, 1960 and March 15, 2021 in MEDLINE, EMBASE, Cochrane Central, OVID, and EBM reviews.

Methods section:
Search strategy
Our proposed literature search strategy is outlined in the supplementary appendix 2. Our planned systematic search will proceed until March 15, 2021.

- Please remove the Conclusions header as this is not required.

This has been removed.